Causality and Machine Learning: *Estimation and Causal Inference*

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Course website:

- https://shorturl.at/efTNq
- https://johandh2o.github.io/causal-inference-summer-school

Quick survey

Please raise your hand if you are familiar with, have used, heard about, or read about:

1. Potential outcomes

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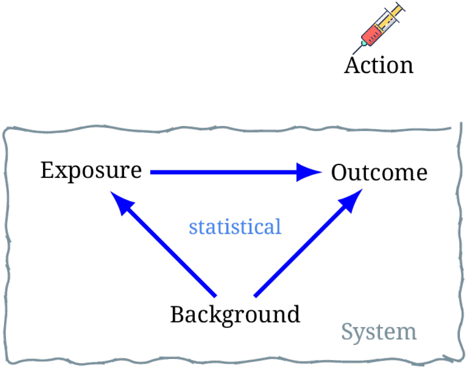
How we will work today

- Please raise your hand or interrupt me at any time if something is unclear.
- The session is designed to be mid-level. If you are interested in introductory or advanced material, I am happy to point you to further resources.
- Some topics will be covered quickly as a panoramic overview. Please contact me if you would like to dive deeper.
- We will use small exercises and labs to keep the session interactive.
- For any extra question, suggestion, or concern, contact me at *johanmd@uio.no*.

1. Causality refresher

Causal inference

Causal inference is the methodological and scientific process of drawing conclusions about the strength of effects induced on some variables in a system (outcomes) by interventions or actions applied to other variables in the same system (treatments / exposures).



Causal vs. statistical inference

Causal inference is fundamentally distinct from statistical inference.

- External actions that break natural relations vs. passive relations as they are observed.
- Changes *ceteris paribus* vs. regular empirical changes (all move, no holding constant).

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Let A be a **treatment** and Y be an **outcome**:

Causal questions

- What would happen to Y if we set $A = a$?
- Would Y have changed if A had been different?
- Which intervention value a for A increases Y the most?

Statistical questions

- How are A and Y associated?
- How well does A predict Y?
- What is the best model to predict Y using A?
- How does Y vary across observed groups X?

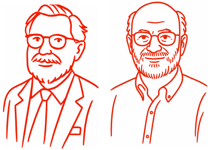
2. Formalisms

Formalisms

Pearl's Structural Causal Model (SCM)

- Primitive: an SCM or a causal graph,
- Identification: graphical conditions (+ positivity),
- Focus: hierarchy (associations $\rightarrow$ interventions $\rightarrow$ counterfactuals),
- Common users: engineers, computer scientists, psychologists...

Neyman–Rubin Causal Model (RCM)



- Primitive: potential outcomes,
- Identification: consistency +positivity +SUTVA +ignorability/exchangeability,
- Focus: randomized experiments for interventions, pseudo-exp, emulations,
- Common users: economists, epidemiologists, biostatisticians...

Potential outcomes

We have data on people in a population P and their treatment use:

- $W \in \mathcal{W}$ is **pre-exposure background** information, e.g. age, gender,...,
- $A \in \{0, 1\}$ is a **binary point exposure**, e.g. 1=medication, 0=control,
- $Y \in \mathcal{Y}$ is a **continuous or discrete outcome**, e.g. clinical state, survival, progression....

Outcomes:

- Y_i is the observed outcome of unit i (under their actual treatment),
- Y_i^a is the potential outcome of unit i had they received treatment $A = a$,
  - Y_i^1 their potential outcome under medication,
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The fundamental problem of causal inference: for every unit i we only observe **one** of the potential outcomes, either if $A_i = 0$ then $Y_i^0 = Y_i$, or if $A_i = 1$ then $Y_i^1 = Y_i$; **never both**.

Consistency: If we see $A_i = a$ then $Y_i^a = Y_i$.

Potential outcomes, notation and SCMs

Many equivalent notations for potential outcomes

$$Y_i^a, \qquad Y_i^{A=a}, \qquad Y_i^{A\leftarrow a}, \quad Y_i(a), \qquad Y^a(u_i).$$

Potential outcomes are central in the RCM, but can be derived from an SCM too.

If $Y = f_Y(W, A, u_Y)$ is the mechanism determining Y in an SCM, then $Y_i^a = Y^a(u_i)$ is the value that Y takes when A is replaced by a in its structural equation for a unit with identity $u_i = (u_{W,i}, u_{A,i}, u_{Y,i})$ (Pearl 2009).

$$Y_i^a = f_Y(W_i^a,\ a,\ u_{Y,i}),$$

where $W_i^a = W_i$ if W is pre-treatment.

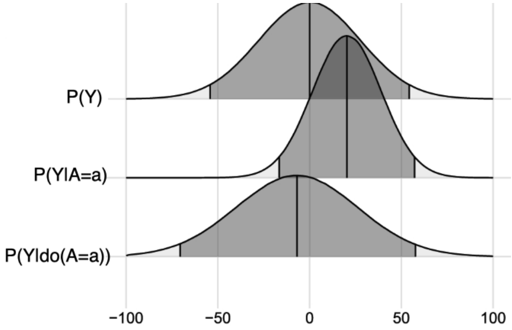
3. Formulation of causal effects, queries and estimands

Causal effects, queries and estimands

(1) We can ask: what is the distribution of an outcome **after** an intervention that sets $A = a$

The (post-)interventional distribution

$$P_Y(y \mid \mathrm{do}(A = a)) \equiv P_{Y^a}(y)$$



Causal effects, queries and estimands

(2) We can ask: what is the **expected outcome after** an intervention that sets $A = a$

The expected potential outcome

$$\mathbb{E}[Y \mid \mathrm{do}(A = a)] \equiv \mathbb{E}[Y^a]$$

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(3) We can ask: what is the **expected outcome after** an intervention that sets $A = a$ for a subpopulation with $X = x$

The expected conditional potential outcome

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Why is $X = x$ not inside a *do*-operator?

Causal effects, queries and estimands

(4) We can ask: what is the **expected difference in outcomes between** two treatments $A = 1$ vs $A = 0$

Average treatment effect (ATE or ACE)

$$\text{ATE} = \mathbb{E}[Y \mid \text{do}(A = 1)] - \mathbb{E}[Y \mid \text{do}(A = 0)] \equiv \mathbb{E}[Y^1 - Y^0]$$

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Conditional ATE (CATE) - Heterogeneous treatment effect (HTE)

$$\text{CATE}(x) = \mathbb{E}[Y \mid \text{do}(A = 1), X = x] - \mathbb{E}[Y \mid \text{do}(A = 0), X = x] \equiv \mathbb{E}[Y^1 - Y^0 \mid X = x]$$

Two caveats to keep in mind about the CATE

Conditional average treatment effect

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Why?

If X is affected by treatment A then we need to use the potential outcomes X^1 and X^0

$$\underbrace{\mathbb{E}[Y^1 - Y^0 \mid Z = z]}_{\text{one population (causal)}} \quad \text{vs} \quad \underbrace{\mathbb{E}[Y^1 \mid X^1 = x] - \mathbb{E}[Y^0 \mid X^0 = x]}_{\text{two populations } (\textbf{not causal})}$$

ATT and ATC: effects for policy-relevant subpopulations

The ATE averages the causal effect over the **full population**. Sometimes, however, the relevant target population is not everyone.

$$\mathrm{ATT} = \mathbb{E}\left[Y^1 - Y^0 \mid A = 1\right]$$

Among those who actually received treatment, what was the average effect of receiving it?

$$\mathrm{ATC} = \mathbb{E}\left[Y^1 - Y^0 \mid A = 0\right]$$

Among those who did not receive treatment, what would be the average effect of receiving it?

These estimands are useful when treatment effects are **heterogeneous** and the scientific or policy question targets a specific group: the currently treated or the currently untreated.

ATE $=$ ATT when

- treatment effects are constant: $Y^1 - Y^0 = \tau$ for everyone;
- treatment assignment is independent of the individual treatment effect: $A \perp\!\!\!\perp (Y^1 - Y^0)$.

Marimo lab

- https://shorturl.at/hO2wM
- https://johandh2o.github.io/causal-inference-summer-school/day3.html

You can interact online (Chrome is better), or download files and run locally. Required packages

```
pip3 install numpy pandas scipy scikit-learn matplotlib statsmodels
jupyterlab marimo pgmpy
```

```
marimo edit /your-path/your-folder
```

4. The causal roadmap

The causal roadmap

System / problem

5. Identification from observational data

Identification

Is it possible to express the causal effect as a **statistical** estimand?
Is it possible to learn the causal effect from **observational / passive data?**

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Both frameworks (SCM/RCM) use graphical criteria for identification, but the RCM focuses more on their consequences in terms of conditional independence using potential outcomes.

Let W be a set of pre-treatment variables:

$Y \perp\!\!\!\perp_d A \mid W$ in graph $\mathcal{G}_{\underline{A}}$

A d-**separation condition** on $\mathcal{G}_{\underline{A}}$ (deleting the arrows out of A), or an application of *do*-**calculus**.

$Y^a \perp\!\!\!\perp A \mid W$ for all $a \in \operatorname{dom} A$

Ignorability/exchangeability condition: given W, treatment assignment is as-good-as random for the potential outcomes.

Example: ATE and the backdoor criterion

Suppose we observe data $O = (W, A, Y) \sim P$:

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W satisfies the **backdoor criterion**, so we have $Y \perp\!\!\!\perp_d A \mid W$ in $\mathcal{G}_{\underline{A}} \Rightarrow$ $Y^a \perp\!\!\!\perp A \mid W$

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$$\text{ATE} = \mathbb{E}[Y^1 - Y^0] = \mathbb{E}_W \mathbb{E}[Y \mid W, A = 1] - \mathbb{E}_W \mathbb{E}[Y \mid W, A = 0]$$

Let's pause and recap

- For two or more values of the treatment (medication vs control), **we never observe both for the same unit** (*the fundamental problem of C.I.*).

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- A common estimand is the ATE. It is typically a number/vector. Another common estimand is the CATE. It is typically a **function**.
- Before estimating a causal effect, we must first ensure that it is **well formulated** and **identified** under explicit assumptions.

Questions?

6. Estimation

Plug-in estimation

We have data of size n, data $= \{O_i\}_{i=1}^n$ with $O_i = (W_i, A_i, Y_i) \overset{iid}{\sim} P$. Let us denote $\psi = \text{ATE}$

$$\psi = \mathbb{E}_W \underbrace{\mathbb{E}[Y \mid W, A = 1]}_{\text{regression}} - \mathbb{E}_W \underbrace{\mathbb{E}[Y \mid W, A = 0]}_{\text{regression}}$$

The expectation outside can be approximated with the average, so an estimator could be:

$$\widehat{\psi} = \frac{1}{n}\sum_{i=1}^{n} \widehat{\mathbb{E}}[Y \mid W_i, A = 1] - \frac{1}{n}\sum_{i=1}^{n} \widehat{\mathbb{E}}[Y \mid W_i, A = 0]$$

How should we fit such a regression model? Linear regression? **ML?**

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Plug-in (ATE/CATE)

1. S-learner
2. T-learner
3. (X-learner)
4. NN architectures

Orthogonal (ATE/CATE*)

1. AIPW
2. One-step / DML
3. TMLE*

Heterogeneous (CATE)

1. R-learner
2. DR-learner
3. Forest-based*

Why are fully parametric models risky in observational causal inference?

linear outcome regression: $\boxed{Y_i = \beta_0 + \beta_1 A_i + \beta_2^\top W_i + \epsilon_i}$, with $\epsilon_i \overset{iid}{\sim} N(0, \sigma^2)$.

In observational studies, causal identification already depends on strong assumptions. Adding rigid parametric models adds more assumptions and can produce **misspecification bias** and **poor inference**.

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- Outcome and treatment assignment are rarely simple.
- Missing interactions or nonlinearities.
- Chosen functions are often unrealistic.
- Selection on high dimensional data is challenging.
- **Identification is nonparametric! Estimation should be as well.**

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overconfidence $\implies$ unreliable inference

- Some strata have little or no overlap: parametric models fill gaps by extrapolating.
- Standard errors are too small, as they may ignore sources of uncertainty.
- Confidence intervals may have undercoverage, **with coverage tending to zero**.

Rather than relying only on rigid parametric models, modern causal inference prefers nonparametric models and ML

Plug-in estimation: S-learner

Based on a **s**ingle outcome regression surface. Recipe:

1. Learn one regression on the whole data using a loss function L and a model class $\mathcal{Q}$:

$$\widehat{Q} = \arg\min_{Q \in \mathcal{Q}} \frac{1}{n} \sum_{i=1}^{n} L\left(Y_i, Q(W_i, A_i)\right).$$

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2. For each unit i, predict the outcome under treatment and under control:

$$\text{two predictions for } i: \qquad \widehat{Q}(W_i, 1) \quad \text{and} \quad \widehat{Q}(W_i, 0).$$

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3. Compute the difference, and average:

$$\widehat{\psi}_{\mathrm{S}} = \frac{1}{n} \sum_{i=1}^{n} \left\{ \widehat{Q}(W_i, 1) - \widehat{Q}(W_i, 0) \right\}.$$

Summary: learn one outcome surface and compare its predictions when A is set to 1 versus 0.

Plug-in estimation: T-learner

Based on **t**wo regression surfaces. Recipe:

1. Learn one outcome regression **for each treatment arm** using loss L and model class $\mathcal{Q}$:

$$\widehat{Q}_1 = \arg\min_{Q \in \mathcal{Q}} \frac{1}{n_1} \sum_{i:A_i=1} L\left(Y_i, Q(W_i)\right),$$

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Summary: learn one outcome surface among treated units and another among controls, then contrast them.

S-learner vs T-learner

S-learner: **advantages**

- Uses all observations to learn one outcome model.
- Simple to implement with any ML algorithm.
- Can be stable when sample sizes are small.

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- Can ignore effect heterogeneity (CATE).

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T-learner: **advantages**

- Allows different outcome models by treatment arm.
- More flexible for heterogeneous effects.
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T-learner: **disadvantages**

- Each model uses only part of the data.
- Can be unstable with imbalanced treatment groups.
- May extrapolate poorly where one arm has little support.

Marimo lab

- https://shorturl.at/hO2wM
- https://johandh2o.github.io/causal-inference-summer-school/day3.html

Lunch time!

But... there are some caveats about the use of data-adaptive ML methods in causal inference

Why is flexible ML plug-in risky in causal inference?

ML outcome plug-in: $\boxed{\widehat{Q} = \arg\min_{Q \in \mathcal{Q}} \frac{1}{n} \sum_{i=1}^{n} \left(Y_i - Q(W_i, A_i)\right)^2}$: min MSE, best BV trade-off.

Flexible ML helps reduce misspecification, but most ML algorithms are designed for **prediction**, not directly for **causal estimation** or **valid inference**.

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prediction loss $\not\Rightarrow$ causal accuracy

- Optimizes prediction of Y, not bias of $\widehat{\psi}$.
- Bias–variance trade-offs may accept too much bias for the target parameter.
- **It only uses a model for Q, which ignores other components such as the treatment model.**.

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black-box fitting $\implies$ hard inference

- Many ML estimators have complex asymptotics.
- Standard errors for $\widehat{\psi}$ are not automatic.
- Overfitting can invalidate inference.
- Uncertainty in nuisance functions is hard to propagate.

Let's pause and recap

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- Parametric models are simple and interpretable, but can be highly biased when their functional form is **misspecified**.
- Machine learning offers flexible plug-in estimation strategies, such as S-, T-, and X-learners.
- These learners involve trade-offs: S-learners may smooth away heterogeneous effects, while T-learners can suffer from instability under small samples or weak overlap.

Let's pause and recap

- Causal estimation should only begin once the target effect is **well formulated** and **identified** under explicit assumptions.
- Parametric models are simple and interpretable, but can be highly biased when their functional form is **misspecified**.
- Machine learning offers flexible plug-in estimation strategies, such as S-, T-, and X-learners.
- These learners involve trade-offs: S-learners may smooth away heterogeneous effects, while T-learners can suffer from instability under small samples or weak overlap.
- The central warning is that prediction and causal inference are not the same: a model can predict well and still estimate causal effects poorly and might not provide valid inference.

Questions?

7. Orthogonal estimation with semiparametric theory

Inverse probability weighting (IPW)

A key equality via **inverse probability weighting (IPW)** or **importance sampling** is:

$$\mathbb{E}_W \mathbb{E}[Y \mid W, A = a] = \mathbb{E}\left[\frac{\mathbb{I}(A = a)Y}{\mathbb{P}(A = a \mid W)}\right],$$

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which is valid under **positivity** $\mathbb{P}(A = a \mid W = w) > 0$ for all w.

Proof.

$$\begin{aligned}
\mathbb{E}_W \mathbb{E}[Y \mid W, A = a] &= \int p(w) \int y\, p(y \mid w, a)\, \mathrm{d}y \mathrm{d}w \\
&= \int p(w) \sum_{a' \in \{0,1\}} \int \mathbb{I}(a' = a)\, y\, p(y \mid w, a')\, \mathrm{d}y \mathrm{d}w \\
&= \int p(w) \sum_{a' \in \{0,1\}} \int \frac{\mathbb{I}(a' = a)}{p(a' \mid w)}\, y\, p(y, a' \mid w)\, \mathrm{d}y \mathrm{d}w \\
&= \mathbb{E}_W \mathbb{E}_{A,Y \mid W}\left[\frac{\mathbb{I}(A = a)Y}{p(a \mid W)}\right] = \mathbb{E}\left[\frac{\mathbb{I}(A = a)Y}{\mathbb{P}(A = a \mid W)}\right].
\end{aligned}$$

AIPW estimation

Plug-in estimators only use a model for the outcome regression Q. Can we leverage a model for A, $\pi(a \mid w) = \mathbb{P}(A = a \mid W = w)$ (**propensity score**) as well?

$$\mathbb{E}[Y^a] = \mathbb{E}_W \mathbb{E}[Y \mid W, A = a] \qquad \text{identification}$$

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&= \mathbb{E}_W \mathbb{E}_{A,Y|W} \left[\frac{\mathbb{I}(A = a)\, (Y - Q(W, A) + Q(W, A))}{\pi(a \mid W)} \right] && \text{add/subtract } Q
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&= \underbrace{\mathbb{E}_W \mathbb{E}_{A,Y|W}\left[\frac{\mathbb{I}(A = a)\,(Y - Q(W, A))}{\pi(a \mid W)}\right]}_{\text{one-step correction}} + \underbrace{\mathbb{E}_W Q(W, a)}_{\text{plug-in}} && \text{IPW}
\end{aligned}
$$

This is the Augmented Inverse-Probability Weighting (AIPW) technique (Robins et al. 1994).

The double robustness property

ATE estimators:

$$\widehat{\psi}_{\text{plug}} = \frac{1}{n}\sum_{i=1}^{n}\left[\widehat{Q}(W_i,1) - \widehat{Q}(W_i,0)\right],$$

$$\widehat{\psi}_{\text{AIPW}} = \frac{1}{n}\sum_{i=1}^{n}\left[\widehat{Q}(W_i,1) - \widehat{Q}(W_i,0) + \left\{\frac{A_i}{\widehat{\pi}(1 \mid W_i)} - \frac{1-A_i}{\widehat{\pi}(0 \mid W_i)}\right\}\{Y_i - \widehat{Q}(W_i,A_i)\}\right].$$

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The plug-in estimator depends only on the outcome model for Q. If $\widehat{Q}$ is misspecified, then $\widehat{\psi}_{\text{plug}}$ is generally biased.

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The plug-in estimator depends only on the outcome model for Q. If $\widehat{Q}$ is misspecified, then $\widehat{\psi}_{\text{plug}}$ is generally biased.

The AIPW estimator combines an outcome model $\widehat{Q}$ and a propensity model $\widehat{\pi}$. It is **doubly robust** (Bang and Robins 2005): it is consistent if either $\widehat{Q}$ or $\widehat{\pi}$ is consistently estimated.

Two chances to remove bias: outcome regression correction + inverse-probability residual correction. DR is one of the most impactful properties in modern causal inference! It is like Actor-Critic in RL.

The influence function

The AIPW estimator is a special case of a broader idea: a one-step corrected estimator using a first-order expansion (von Mises 1947).

Ordinary Taylor expansion

$$f(\widehat{x}) \approx f(x) + (\widehat{x} - x)\, f'(x)\,.$$

Statistical von Mises expansion

$$\psi(\widehat{P}) \approx \psi(P) + (\widehat{\mathbb{E}} - \mathbb{E})\, D_P(O)\,.$$

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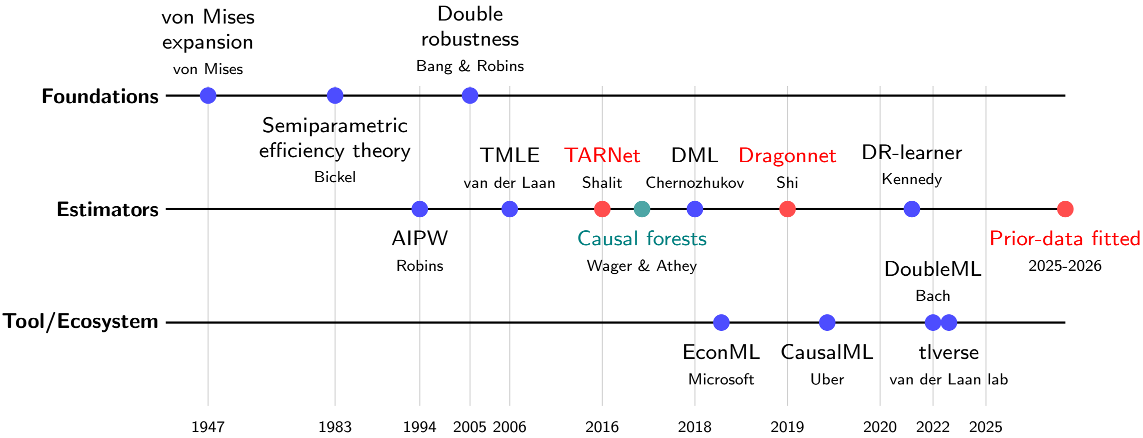
Here D_P is the **influence function**: the first-order derivative of the causal estimand with respect to changes in the distribution of the data $O = (W, A, Y)$.

For the ATE it gives:

$$D_P(O) = \overbrace{\underbrace{\left\{\frac{A}{\pi(1 \mid W)} - \frac{1-A}{\pi(0 \mid W)}\right\}\{Y - Q(W,A)\}}_{\text{one-step/first-order correction}} + \underbrace{Q(W,1) - Q(W,0)}_{\text{plug-in}}}^{\text{the score } \varphi(O)} - \underbrace{\psi}_{\text{ATE}}$$

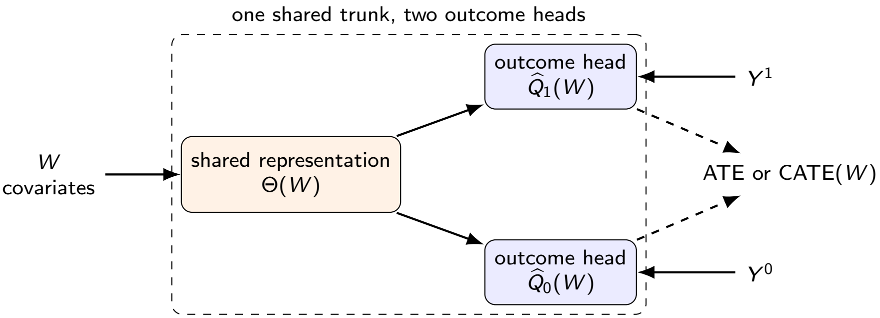
Property: $\mathbb{E}D_P(O) = 0$. Move ψ to the other side and one gets the AIPW!

A schematic timeline of semiparametric theory + causal ML



Methods based on semiparametric theory (influence functions) are dominant in today's causal ML, while emerging directions using neural networks and foundation models are on the rise.

Meta-learning with neural nets: TARNet



Features (Shalit et al. 2017)

- Learns a shared representation $\Theta(W)$.
- Separate outcome heads: treated and control.
- Closer to a T-learner.

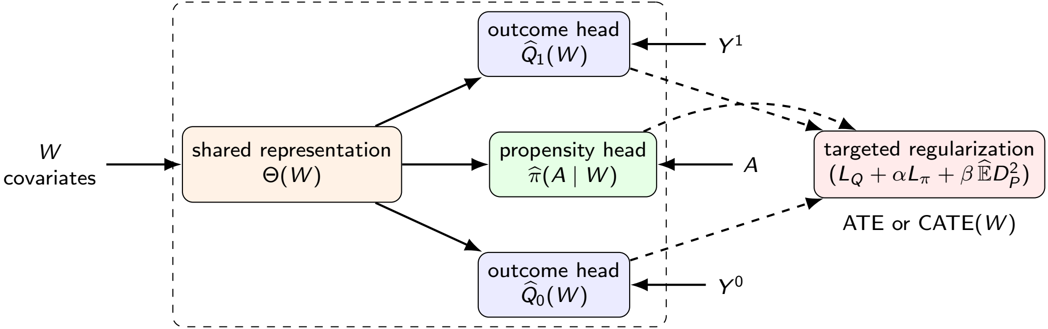
Advantages

- Shares information across arms through the common trunk (like an S-learner).
- Often more stable than fitting two unrelated deep nets.

Disadvantages

- Prediction-oriented.
- Does not explicitly model the propensity score.
- **No DR, no automatic inference**.
- Biased under poor overlap.

Meta-learning with neural nets: Dragonnet



Features (Shi et al. 2019)

- TARNet + propensity head.
- Learns $\widehat{Q}_1(w)$, $\widehat{Q}_0(w)$, and $\widehat{\pi}(a \mid w)$ jointly.
- Includes targeted regularization inspired by semiparametric ideas.

Advantages

- Better aligned with causal adjustment than pure prediction nets.
- Uses both outcome and treatment-assignment information.

Disadvantages

- Complex to tune / train.
- **No DR no inference.**
- Harder to interpret than simpler semiparametric estimators.

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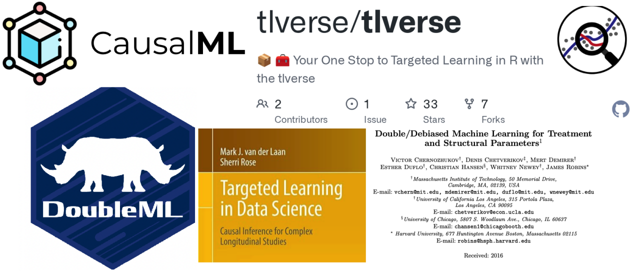
- https://shorturl.at/hO2wM
- https://johandh2o.github.io/causal-inference-summer-school/day3.html

State of the art: orthogonal estimation



Orthogonal estimators are now central

- Combine flexible ML with semiparametric theory.
- Often provide double robustness.
- Valid asymptotic inference after ML nuisance estimation.
- Have become highly visible in statistics, econometrics, ML, and AI venues.

But there are important challenges

- Mathematically advanced, with a steep learning curve.
- Require careful implementation: cross-fitting, nuisance estimation, positivity checks.
- Less plug-and-play than standard predictive ML workflows.

DML and one-step estimation

DML is essentially AIPW / one-step estimation with two extra ingredients: **flexible ML** for nuisance functions and **cross-fitting** to avoid using the same data twice (Chernozhukov et al. 2018).

$$Q(w, a) = \mathbb{E}[Y \mid W = w, A = a], \qquad \pi(a \mid w) = \mathbb{P}(A = a \mid W = w).$$

For unit i in fold $k(i)$, estimate them on the other folds: $\widehat{\eta}_{-k(i)} = \left(\widehat{Q}_{-k(i)}, \widehat{\pi}_{-k(i)}\right)$.

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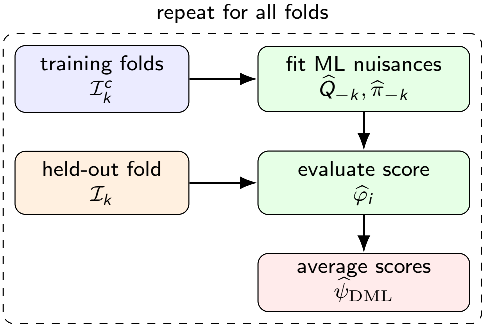
For unit i in fold $k(i)$, estimate them on the other folds: $\widehat{\eta}_{-k(i)} = \left(\widehat{Q}_{-k(i)}, \widehat{\pi}_{-k(i)}\right)$.
Then evaluate the AIPW score out-of-sample:

$$\begin{aligned}\widehat{\varphi}_i = {} & \widehat{Q}_{-k(i)}(W_i,1) - \widehat{Q}_{-k(i)}(W_i,0) \\ & + \left\{\frac{A_i}{\widehat{\pi}_{-k(i)}(1 \mid W_i)} - \frac{1-A_i}{\widehat{\pi}_{-k(i)}(0 \mid W_i)}\right\} \\ & \times \{Y_i - \widehat{Q}_{-k(i)}(W_i, A_i)\}.\end{aligned}$$

$$\widehat{\psi}_{\mathrm{DML}} = \frac{1}{n}\sum_{i=1}^{n} \widehat{\varphi}_i$$



Why is DML debiased?

Reorder terms in the expansion:

$$\widehat{\psi}_{\mathrm{DML}} - \psi = \underbrace{(\widehat{\mathbb{E}} - \mathbb{E}) D_P(O)}_{\text{sampling noise}} + \underbrace{R_1(\widehat{P}, P)}_{\text{1-order bias}} + \underbrace{R_2(\widehat{P}, P)}_{\text{2-order remainder}} + o_P(n^{-1/2})$$

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[1] The orthogonal scores in DML **force the 1-order bias to be zero**

$$R_1(\widehat{P}, P) := \frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}\Big[\underbrace{\psi(P_t) + D_{P_t}(O)}_{\text{the score } \varphi_t(O)}\Big]_{t=0} = 0, \quad P_t = P + t(\widehat{P} - P).$$

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[2] If models converge fast enough, $\|\widehat{Q} - Q\| = o_P(n^{-1/4})$ and $\|\widehat{\pi} - \pi\| = o_P(n^{-1/4})$, one has

$$|R_2(\widehat{P}, P)| \lesssim \|\widehat{Q} - Q\| \, \|\widehat{\pi} - \pi\| = o_P(n^{-1/2}).$$

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[3] Cross-fitting controls the empirical fluctuation from using $D_{\widehat{P}}$:

$$(\widehat{\mathbb{E}} - \mathbb{E})D_{\widehat{P}}(O) = (\widehat{\mathbb{E}} - \mathbb{E})D_P(O) + o_P(n^{-1/2}).$$

Why is DML debiased?

Reorder terms in the expansion:

$$\widehat{\psi}_{\mathrm{DML}} - \psi = \underbrace{(\widehat{\mathbb{E}} - \mathbb{E})D_P(O)}_{\text{sampling noise}} + \underbrace{R_1(\widehat{P}, P)}_{\text{1-order bias}} + \underbrace{R_2(\widehat{P}, P)}_{\text{2-order remainder}} + o_P(n^{-1/2})$$

[1] The orthogonal scores in DML **force the 1-order bias to be zero**

$$R_1(\widehat{P}, P) := \frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}\Big[\underbrace{\psi(P_t) + D_{P_t}(O)}_{\text{the score } \varphi_t(O)}\Big]_{t=0} = 0, \quad P_t = P + t(\widehat{P} - P).$$

[2] If models converge fast enough, $\|\widehat{Q} - Q\| = o_P(n^{-1/4})$ and $\|\widehat{\pi} - \pi\| = o_P(n^{-1/4})$, one has

$$|R_2(\widehat{P}, P)| \lesssim \|\widehat{Q} - Q\| \, \|\widehat{\pi} - \pi\| = o_P(n^{-1/2}).$$

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We get an asymptotic normal distribution

$$\widehat{\psi}_{\mathrm{DML}} - \psi = (\widehat{\mathbb{E}} - \mathbb{E})D_{\widehat{P}}(O) + o_P(n^{-1/2}) \Rightarrow \boxed{\widehat{\psi}_{\mathrm{DML}} \approx N\left(\psi,\ \sigma^2_{\mathrm{DML}} = n^{-1}\mathbb{E}D^2_{\widehat{P}}(O)\right)}$$

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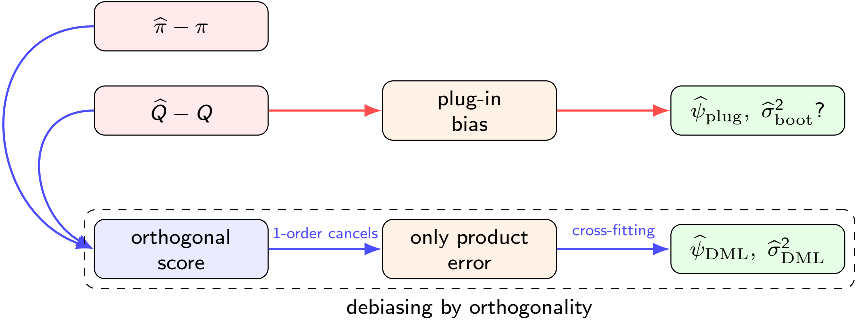

DML has become popular because it combines several attractive properties:

- it allows the use of flexible machine learning methods;
- it can deliver double robustness;
- it provides a principled route to confidence intervals.

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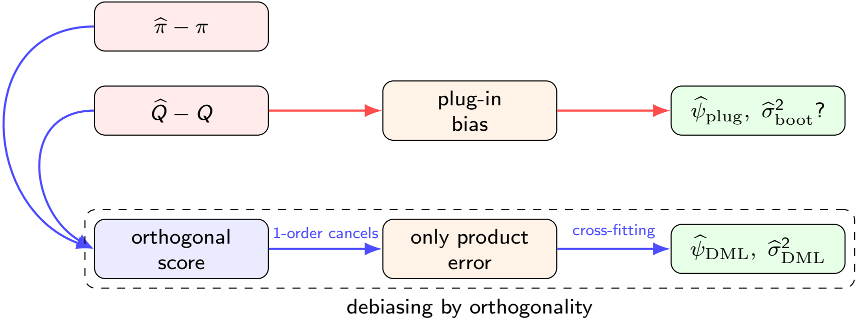

DML has become popular because it combines several attractive properties:

- it allows the use of flexible machine learning methods;
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- it provides a principled route to confidence intervals.

But... suppose we are estimating **a probability**. The plug-in estimate is 0.95, but the one-step correction moves it by 0.06. What should the final estimate be?

Targeted Minimum-Loss Estimation (TMLE)

TMLE starts with flexible ML, but then targets the initial regression toward the estimand ψ (van der Laan and Rose 2011, 2018).

$$\widehat{\psi}_{\text{plug}} = \frac{1}{n}\sum_{i=1}^{n}\left[\widehat{Q}^{0}(W_i, 1) - \widehat{Q}^{0}(W_i, 0)\right].$$

$$\widehat{\psi}_{\text{TMLE}} = \frac{1}{n}\sum_{i=1}^{n}\left[\widehat{Q}^{\star}(W_i, 1) - \widehat{Q}^{\star}(W_i, 0)\right].$$

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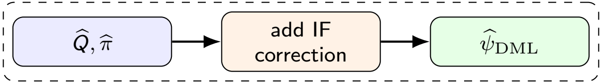
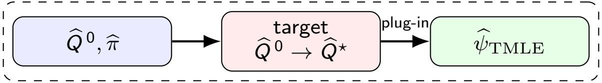
DML / one-step



targeted plug-in



TMLE respects the natural bounds of the parameters, but it is asympt. equivalent to DML.

TMLE: the targeting step

Remember the influence function

$$D_P(O) = \overbrace{\underbrace{\left\{\frac{A}{\pi(1 \mid W)} - \frac{1-A}{\pi(0 \mid W)}\right\}\{Y - Q(W,A)\}}_{\text{one-step/first-order correction}} + \underbrace{Q(W,1) - Q(W,0)}_{\text{plug-in}}}^{\text{the score } \varphi(O)} - \underbrace{\psi}_{\text{ATE}}$$

The red part is called the clever covariate or Riesz representer of the ATE:

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The **targeting step updates the initial outcome regression** $\widehat{Q}^0$ in the direction of the clever covariate. With an L^2-loss, this is the one-dimensional least-squares fluctuation:

$$\widehat{\epsilon} = \frac{\sum_{i=1}^n H_{\widehat{\pi}}(W_i, A_i)\left\{Y_i - \widehat{Q}^0(W_i, A_i)\right\}}{\sum_{i=1}^n H_{\widehat{\pi}}(W_i, A_i)^2}$$

OLS coeff.: residuals $\sim H$ with no intercept,

$$\widehat{Q}^\star(W,A) = \widehat{Q}^0(W,A) + \widehat{\epsilon}\, H_{\widehat{\pi}}(W,A).$$

This makes the one-step correction approx. zero. Other loss functions produce other updates.

Let's pause and recap

- Blindly using ML for causal estimation can produce biased plug-in estimates: good prediction does not guarantee valid causal inference. This is why **debiased** and **double-robust** estimators are desirable.

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- Blindly using ML for causal estimation can produce biased plug-in estimates: good prediction does not guarantee valid causal inference. This is why **debiased** and **double-robust** estimators are desirable.
- AIPW combines an outcome model $\widehat{Q}$ and a treatment model $\widehat{\pi}$, giving **double robustness**: consistency can hold if either nuisance model is correct.
- DML and one-step estimators use ML for nuisance functions, but rely on **orthogonality** and **cross-fitting** to control overfitting and nuisance bias.
- TMLE also uses the influence function, but instead of only correcting $\widehat{\psi}$, it **targets** $\widehat{Q}$ so the final plug-in estimate respects the parameter space.

Questions?

Marimo lab

- https://shorturl.at/hO2wM
- https://johandh2o.github.io/causal-inference-summer-school/day3.html

8. Heterogeneous treatment effect estimation

Heterogeneous treatment effects

We keep the same observational data structure:

$$O = (W, A, Y) \sim P, \qquad A \in \{0, 1\}, \qquad W \text{ pre-treatment.}$$

$$\boxed{\text{ATE} = \psi := \mathbb{E}\left[Y^1 - Y^0\right] = \mathbb{E}_W[\tau(W)]}, \qquad \boxed{\text{CATE} = \tau(w) := \mathbb{E}\left[Y^1 - Y^0 \mid W = w\right]}$$

Ignorability + Positivity: $\boxed{\tau(w) = Q(w, 1) - Q(w, 0)}$, $\quad Q(w, a) := \mathbb{E}[Y \mid W = w, A = a]$.

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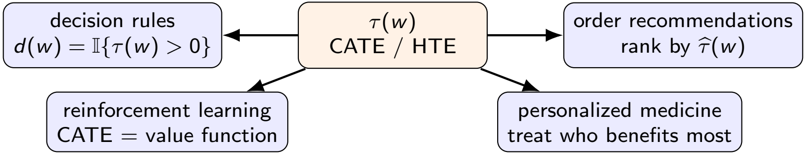
The ATE answers "does treatment work on average?" whereas the CATE asks for whom and under which context does treatment work? (Künzel et al. 2019; Wager and Athey 2018).
It is a very important object for decision-making.



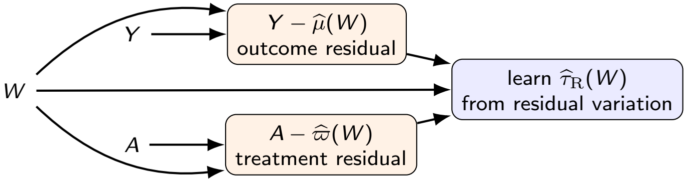
R-learner: remove the parts that are easy to predict

Define two nuisance functions:

$$\mu(w) := \mathbb{E}[Y \mid W = w], \qquad \varpi(w) := \mathbb{P}(A = 1 \mid W = w) \equiv \pi(1 \mid w).$$

The key **Robinson residualization** (Robinson 1988) is

$$Y - \mu(W) = \{A - \varpi(W)\}\, \tau(W) + \varepsilon, \qquad \mathbb{E}[\{A - \varpi(W)\}\varepsilon \mid W] = 0.$$



Intuition: first remove the part of Y and A explained by W; what remains is the causal contrast signal (Nie and Wager 2021).

R-learner: recipe

A practical recipe:

1. Split the data or use cross-fitting.

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3. Construct the residualized pseudo-regression problem:

$$\widetilde{Y}_i := \frac{Y_i - \widehat{\mu}(W_i)}{A_i - \widehat{\varpi}(W_i)}, \qquad \widehat{\omega}_i := \{A_i - \widehat{\varpi}(W_i)\}^2.$$

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What to watch?

- Not doubly robust in the AIPW sense.
- Poor overlap creates instability.

DR-learner: turn CATE estimation into regression

The DR-learner builds a doubly robust pseudo-outcome for the CATE. Estimate

$$Q(w, a) = \mathbb{E}[Y \mid W = w, A = a], \qquad \pi(a \mid w) = \mathbb{P}(A = a \mid W = w).$$

For each observation, use the **AIPW / DML score** as the **pseudo-outcome**:

$$\widehat{\varphi}_i := \widehat{Q}(W_i, 1) - \widehat{Q}(W_i, 0) + \left\{ \frac{A_i}{\widehat{\pi}(1 \mid W_i)} - \frac{1 - A_i}{\widehat{\pi}(0 \mid W_i)} \right\} \{Y_i - \widehat{Q}(W_i, A_i)\}.$$

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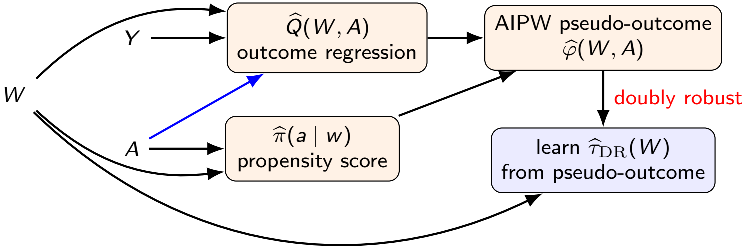
Then estimate the CATE by regressing this pseudo-outcome on W (Kennedy 2023):

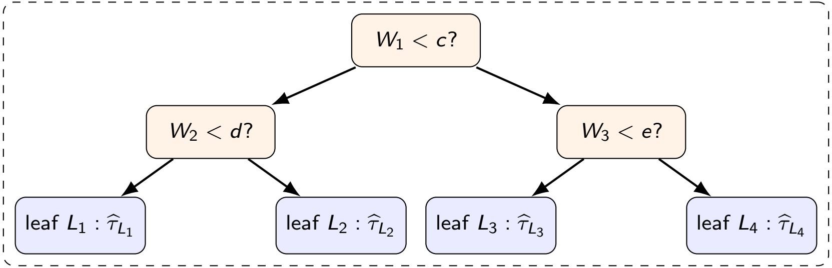
$$\widehat{\tau}_{\mathrm{DR}} \in \arg\min_{\tau \in \mathcal{T}} \frac{1}{n} \sum_{i=1}^{n} \{\widehat{\varphi}_i - \tau(W_i)\}^2.$$



Causal forests: trees for treatment effects

A standard regression tree splits covariate space to improve prediction of Y.
A causal tree / forest splits covariate space to find regions with different causal effects.

splits chosen to reveal heterogeneity in $\tau(w)$



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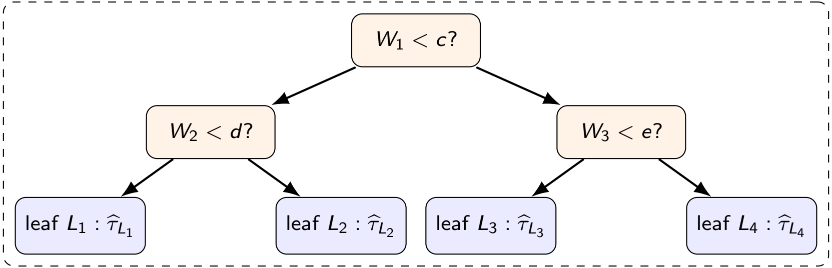


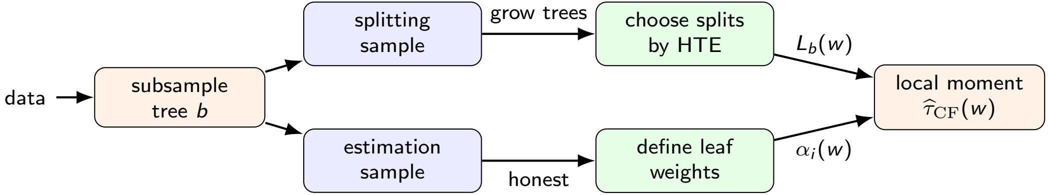
Inside a leaf L, in a randomized experiment one could estimate

$$\widehat{\tau}_L = \frac{1}{n_{1,L}} \sum_{i:W_i \in L, A_i = 1} Y_i - \frac{1}{n_{0,L}} \sum_{i:W_i \in L, A_i = 0} Y_i .$$

In observational studies, causal forests use **sample splitting and orthogonal local moments** (Athey and Imbens 2016; Wager and Athey 2018)

Causal forests: learning algorithm

A causal forest learns many causal trees L_b. The weights $\alpha_i(w)$ are then induced by how often unit i falls in the same leaf as w.



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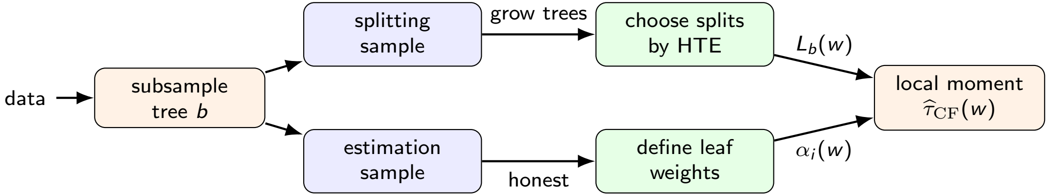

Splits are chosen to create leaves with different causal effects. Given a target point w, let $L_b(w)$ be the leaf containing w in tree b. With honesty, weights are computed as:

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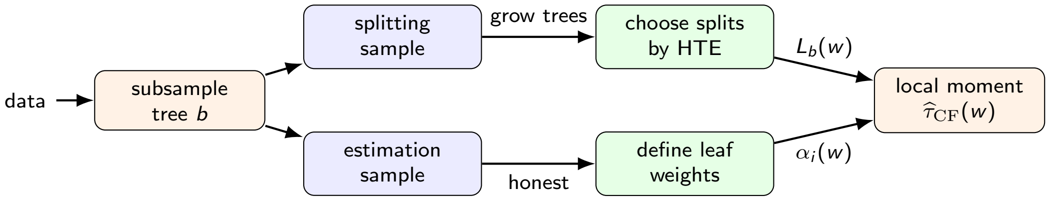

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These learned weights define a local R-learner:

$$\widehat{\tau}_{\mathrm{CF}}(w) = \frac{\sum_{i=1}^{n} \alpha_i(w)\{A_i - \widehat{\varpi}(W_i)\}\{Y_i - \widehat{\mu}(W_i)\}}{\sum_{i=1}^{n} \alpha_i(w)\{A_i - \widehat{\varpi}(W_i)\}^2}.$$

Comparing HTE learners

All these methods estimate the same **function**,

$$\boxed{\tau(w) = \mathbb{E}[Y^1 - Y^0 \mid W = w] \overset{id}{=} Q(w,1) - Q(w,0)},$$

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- **Causal forests** learn adaptive neighborhoods around w and estimate local orthogonal effects. They are powerful for discovering heterogeneity, but more complicated and still not doubly robust.

Marimo lab

- https://shorturl.at/hO2wM
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9. Final recap and conclusions

Final recap: ML for causal estimation

- **Causal estimation starts before ML.** Before estimating any effect, we need a well-defined causal estimand, identification assumptions, and a statistical estimand:

$$\text{causal question} \longrightarrow \text{identification} \longrightarrow \text{estimation.}$$

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- **HTE estimation moves from average effects to decisions.** Estimating $\tau(w) = \mathbb{E}[Y^1 - Y^0 \mid W = w]$ supports targeting, recommendations, personalized medicine, and policy learning, but requires stronger local overlap and more careful validation.

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- **HTE estimation moves from average effects to decisions.** Estimating $\tau(w) = \mathbb{E}[Y^1 - Y^0 \mid W = w]$ supports targeting, recommendations, personalized medicine, and policy learning, but requires stronger local overlap and more careful validation.

- **Many challenges remain.** Modern causal ML still faces difficult problems: continuous treatments, inference for continuous CATEs, missing data, selection bias, interference, and reliable uncertainty quantification with adaptive ML.

Final recap: ML for causal estimation

- **Causal estimation starts before ML.** Before estimating any effect, we need a well-defined causal estimand, identification assumptions, and a statistical estimand:

$$\text{causal question} \longrightarrow \text{identification} \longrightarrow \text{estimation.}$$

- **ML is useful because nuisance functions are hard.** Flexible models can estimate objects such as $Q(w, a)$, $\varpi(w)$, $\mu(w)$, and $\tau(w)$ without imposing strong parametric structure.

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- **HTE estimation moves from average effects to decisions.** Estimating $\tau(w) = \mathbb{E}[Y^1 - Y^0 \mid W = w]$ supports targeting, recommendations, personalized medicine, and policy learning, but requires stronger local overlap and more careful validation.

- **Many challenges remain.** Modern causal ML still faces difficult problems: continuous treatments, inference for continuous CATEs, missing data, selection bias, interference, and reliable uncertainty quantification with adaptive ML.

Contact me if you want to make your final project about **estimation of causal effects with ML** *(`johanmd@uio.no`). Thank you!*

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10. Extra slides

Other common causal estimands

- **Causal risk difference or ratio** for binary outcomes:

$$\mathrm{RD} = \mathbb{P}(Y^1 = 1) - \mathbb{P}(Y^0 = 1), \qquad \mathrm{RR} = \frac{\mathbb{P}(Y^1 = 1)}{\mathbb{P}(Y^0 = 1)}.$$

- **Causal odds ratio** for binary outcomes:

$$\mathrm{OR} = \frac{\mathbb{P}(Y^1 = 1)/\mathbb{P}(Y^1 = 0)}{\mathbb{P}(Y^0 = 1)/\mathbb{P}(Y^0 = 0)}.$$

- **Local average treatment effect** for compliers, using an instrument Z:

$$\mathrm{LATE} = \mathbb{E}\left[Y^1 - Y^0 \mid A^1 > A^0\right].$$

- **Controlled direct effect** fixing a mediator do$(A = 1, M = m)$ vs do$(A = 0, M = m)$:

$$\mathrm{CDE}(m) = \mathbb{E}\left[Y^{1,m} - Y^{0,m}\right].$$

- **Natural direct and indirect effects**:

$$\mathrm{NDE} = \mathbb{E}\left[Y^{1,M^0} - Y^{0,M^0}\right], \qquad \mathrm{NIE} = \mathbb{E}\left[Y^{1,M^1} - Y^{1,M^0}\right].$$

Plug-in estimation: X-learner

Based on imputed individual treatment effects. Recipe:

1. First learn two outcome regressions, as in the T-learner:

$$\widehat{Q}_1(w) = \widehat{\mathbb{E}}[Y \mid W = w, A = 1], \qquad \widehat{Q}_0(w) = \widehat{\mathbb{E}}[Y \mid W = w, A = 0].$$

2. Impute the missing treatment effect for each observed unit:

$$\begin{aligned} \widehat{D}_i^1 &= Y_i - \widehat{Q}_0(W_i), \qquad && A_i = 1, \\ \widehat{D}_i^0 &= \widehat{Q}_1(W_i) - Y_i, \qquad && A_i = 0. \end{aligned}$$

3. Learn two treatment-effect regressions:

$$\widehat{\tau}_1(w) = \widehat{\mathbb{E}}[\widehat{D}^1 \mid W = w, A = 1], \qquad \widehat{\tau}_0(w) = \widehat{\mathbb{E}}[\widehat{D}^0 \mid W = w, A = 0].$$

4. Combine them using the estimated propensity score $\widehat{\varpi}(w) = \widehat{\mathbb{P}}(A = 1 \mid W = w)$, and average:

$$\widehat{\psi}_X = \frac{1}{n} \sum_{i=1}^{n} \left[\widehat{\varpi}(W_i)\widehat{\tau}_0(W_i) + \{1 - \widehat{\varpi}(W_i)\}\widehat{\tau}_1(W_i)\right].$$

Intuition: first impute the missing counterfactual outcomes, then learn and combine two treatment-effect models.

TMLE: targeted minimum-loss estimation

TMLE starts with flexible ML, but then targets the initial regression toward the estimand ψ.

Start with initial estimates

$$\widehat{Q}^{\,0}(W, A), \qquad \widehat{\pi}(A \mid W).$$

Fluctuate the initial outcome regression, and regress **with no intercept** against the clever covariates

$$Y \sim \widehat{Q}^{\,0}(W, A) + \epsilon H_{\widehat{\pi}}(W, A).$$

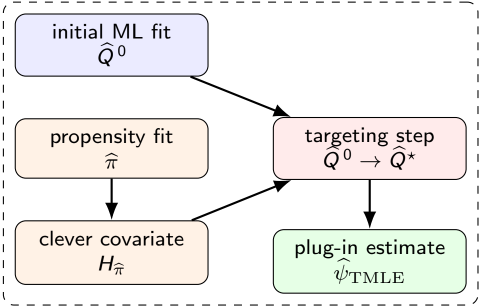
Estimate ϵ by OLS or by **minimizing a loss**

$$\widehat{\epsilon} = \arg\min_{\epsilon} \frac{1}{n} \sum_{i=1}^{n} L\left\{ Y_i, \widehat{Q}^{\,\epsilon}(W_i, A_i) \right\}.$$

Finally set $\text{link}\,\widehat{Q}^{\star} = \text{link}\,\widehat{Q}^{\,0} + \widehat{\epsilon} H_{\widehat{\pi}}$ and plug in:

$$\widehat{\psi}_{\mathrm{TMLE}} = \frac{1}{n} \sum_{i=1}^{n} \left[\widehat{Q}^{\star}(W_i, 1) - \widehat{Q}^{\star}(W_i, 0) \right].$$

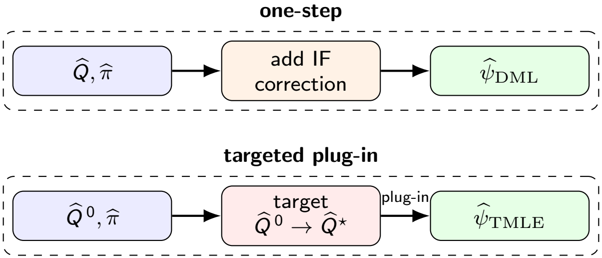
TMLE = ML + causal targeting



The update is usually low-dimensional. We do not refit the whole ML model. We only move $\widehat{Q}^{\,0}$ in the direction that matters for the influence function.

What is the targeting step doing?

DML and TMLE use essentially the same influence-function idea, but implement it differently.



DML / one-step: add the correction after fitting

$$\widehat{\psi}_{\mathrm{DML}} = \widehat{\mathbb{E}}\left[\widehat{Q}(W,1) - \widehat{Q}(W,0)\right] + \widehat{\mathbb{E}}\left[H_{\widehat{\pi}}(A,W)\{Y - \widehat{Q}(W,A)\}\right].$$

TMLE: update the fit until the correction is zero

$$\widehat{\psi}_{\mathrm{TMLE}} = \widehat{\mathbb{E}}\left[\widehat{Q}^{\star}(W,1) - \widehat{Q}^{\star}(W,0)\right] \text{ with } \widehat{\mathbb{E}}\left[H_{\widehat{\pi}}(A,W)\{Y - \widehat{Q}^{\star}(W,A)\}\right] \approx 0.$$

substitution estimator respects bounds asymptotically equivalent

DML corrects $\widehat{\psi}$; TMLE corrects $\widehat{Q}$